

Review

The accessibility of digital technologies for people with visual impairment and blindness: a scoping review

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Abstract

This scoping review aimed to improve the understanding of important factors in digital accessibility for people with visual impairment and blindness, focusing on the first-hand experiences and challenges faced by this target group while using digital technologies. Keywords related to ‘digital technologies,’ ‘accessibility,’ ‘visual impairment,’ and ‘blindness’ were used in searching two databases (n = 683), with additional articles identified by means of manual searches (n = 60). Two reviewers independently screened the titles and abstracts to select 97 articles for full-text screening, of which 49 articles met the inclusion criteria and were selected for review based on the WCAG guidelines, highlighting details for consideration and improvement of the guidelines. The analysis revealed that users suffered from inaccessibility in several ways. For example, many applications or websites are developed for sighted users, where information is communicated through visual content without providing alternatives for assistive technology users. In addition, the lack of keyboard accessibility, shortcuts, or compatibility with different assistive technologies remains a consistent challenge. Furthermore, it was highlighted that simple accessibility measures are not followed adequately or consistently, such as providing alternative text for images or labels for links and buttons. This review highlighted the challenges and consequences of the inaccessibility of digital technologies, providing a detailed explanation regarding the elements that should be considered in the development of digital technologies. It is recommended that people with disabilities should be involved in the design of technology and its accessibility assessment.

Keywords Accessibility · Barriers · Digital technologies · Blind · Visual impairment · Review

1 Introduction

Increasing digitalization brings with it numerous advantages in terms of information procurement, networking, communication, and efficiency, which have revolutionized the everyday and business worlds and driven innovation. While the digital age is constantly opening new doors, speeding up processes, and enabling communication on a global level, the questions of whether this progress is equally accessible to all people, who benefits from this progress, and who is excluded by barriers are now more relevant than ever. This highlights the relevance of digital accessibility, where people with disabilities should also be able to use digital technologies, including hardware as

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well as software [1], without difficulties and barriers [2–4]. Other potential barriers to accessibility are emerging as the digital world evolves [5]; thus, other than a universal design, guidelines could provide a unified framework for the development of accessible digital technologies. For example, the Web Content Accessibility Guidelines (WCAG) are an international standard for the accessible website design of the World Wide Web Consortium (W3C). The guidelines are categorized based on the POUR principles: perceivability, operability, understandability, and robustness. Perceivability aims to ensure that all information, whether in a textual, image or multimedia format, is perceptible to users. Operability describes the need for all users to use the navigation and user interface effectively, including interaction with the keyboard, mouse, and other input devices. Understandability indicates that digital content should be clear and understandable to all users. Robustness aims to ensure that digital content is sufficiently robust to be interpreted by different assistive technologies (AT) (see [6]). It should be noted, however, that the WCAG are a work in progress—there remains room for improvement in the guidelines [7].

Although digital technologies such as smartphones, apps and websites have opened new pathways of inclusion for visually impaired and blind people, many aspects of accessibility are ignored in their development [8]. In our qualitative pre-study [9], we investigated the positive and negative user experiences, barriers, challenges, and opportunities of video conference tools faced by people with visual impairment and blindness. The challenges faced by the target group in using these platforms were enormous, causing not only disruption to their work or university activities but also emotional and psychosomatic problems. In a study to evaluate the accessibility of e-commerce websites for blind users based on the WCAG, Gonçalves et al. [10] reported that blind users found it difficult to complete the tasks and conclude a purchase because the process was complicated and the website was not sufficiently accessible. Regarding alternative text (alt text) for images, Crane et al. [11] studied more than 200 journals based on the WCAG to assess the accessibility of their images to visually impaired and blind users. However, they used an automated accessibility evaluation tool to assess accessibility, rather than the target group, to show that, as expected, not all journals provided alt text for the images. In total, 14.8% of cases had no alt text, 9.6% of cases had insufficient alt text, and none of the journals provided a full interpretation of the meaning of the images. Khan and Khusro [12] reviewed the use of smartphones as ATs and their potential risks and challenges for the target group, highlighting the challenges of touchscreens (e.g., difficulties with text input or a lack of keyboards and buttons), the incompatibility of devices with other ATs, and crowded screens. Tsatsou [13] examined the role of accessibility in the social inclusion and stigmatization of people with disabilities, concluding that accessible digital technologies make them feel part of society, not separate from it [13]. Moreover, studies have highlighted that barriers to digital technologies experienced by visually impaired or blind people hinder their achievements at different stages of their lives, for example, in education [5, 14] or employment [15].

In a systematic review on the subject of the usability of mobile applications for the target group, Al-Razgan et al. [16] summarized that studies on accessibility had mainly focused on their accessibility assessments, challenges faced by the target group, and suggestions for improvement. Senjam et al. [17] expanded on this by focusing on smartphones and apps in their literature review for the target group, concluding that guidelines for developers and information in this regard for people in contact with the target group, e.g., parents, teachers, or medical staff, would be a valuable addition to the field. Oh et al. [18] conducted a systematic review of the accessibility of images for screen reader users when using touchscreens, focusing on the methods used to understand whether any types of images, i.e., with and without motion, were investigated. They concluded that most studies had not involved the target group in the design process, where their input as AT users could play an important role.

In conclusion, although many studies focus on the accessibility of digital technologies for people with visual impairment and blindness and suggest improvements, there is no comprehensive overview of the shortcomings and challenges. Furthermore, based on the literature as well as suggestions for future research, a gap was identified in which an accessible and understandable overview of important accessibility factors for all stakeholders is needed. To the best of our knowledge, no scoping review has evaluated the accessibility of a wide range of digital technologies for blind and visually impaired users based on the POUR principles of the WCAG using first-hand user experience. Therefore, given the importance of digital accessibility for this target group, a scoping review was conducted to explore this gap and investigate the challenges, issues or facilitators in the use of digital technologies by visually impaired and blind people. The review focuses on detailed factors of digital accessibility by extending the POUR items specifically tailored for this target group.

2 Materials and methods

This scoping review was performed according to the PRISMA guidelines for scoping reviews (PRISMA-ScR) [19] as well as the framework proposed by Arksey and O'Malley [20] in five phases:

2.1 Identifying the research question

The purpose of this review was to understand the experiences of people with visual impairment and blindness in using digital technologies, as well as the challenges faced by the target group due to the inaccessibility of digital technologies. Furthermore, it aimed to investigate detailed factors that support better accessibility of digital technologies for the target group and to explore whether following the existing guidelines alone is sufficient for the development of accessible digital technologies.

2.2 Identifying relevant studies

Relevant keywords were selected and pilot-tested for use in two databases, namely Web of Science and PsycInfo. They were the combination of different terms in English and German for “visual impairment” and “blindness” AND “digital technology” AND “accessibility” used in conjunction with truncators such as asterisks (*) and Boolean operators (AND, OR and NOT) depending on the database (Table 1). It is noteworthy that digital technology was used as an umbrella term for this review, covering both hardware, such as smartphones and tablets, and software, such as apps and websites [1]. The search results were exported to Rayyan [21] for further analysis. Additionally, several databases (ACM, IEEE, Google Scholar, EbscoHost, ResearchGate, and the journal of “technology and disability”) were also manually searched and recently published articles were also considered. Articles published in English and German were reviewed. Since technologies change, improve and evolve rapidly, this review intended to identify patterns or issues that are consistent over a broad time period. Therefore, no publication timeframe was considered. Moreover, the context of the studies (e.g., focused only on work or education) was not limited, as the intention was to investigate the first-hand experiences of the target group in general as well as the important factors of accessibility from various perspectives.

2.2.1 Inclusion criteria

Studies were included if they focused on the experiences of people with visual impairment and blindness in using websites, applications, software, educational platforms, etc., the accessibility of digital technologies, ways of using these technologies, and challenges in using these technologies (as reported by the target group).

2.2.2 Exclusion criteria

The following articles were excluded:

- Doctoral theses (as the full text was not available in most cases).
- Usability studies without a focus on accessibility.
- Studies focused on prototypes without reporting their accessibility aspects.

Table 1 An example of the search strings used for the review

Disability	Blind OR “blind person” OR “visually impaired” OR “visual impairments” OR blindness OR “blind people” OR “Visually Disabled” OR “seeing disability” OR “Sehbehinderung” OR Blindheit OR Sehbeeinträchtigung
Technology	“smart devices” OR “digital technology” OR “internet of things” OR “assistive technology” OR “assistive device” OR “digital assistive technology” OR app OR “mobile application” OR software OR tablet OR smartphone OR smartwatch OR “video conference” OR “Web conference” OR “digitale Technologie” OR “assistive Technologie” OR “Webkonferenz” OR Videokonferenz OR “Videoconferencing Tools”
Accessibility	accessibility OR usability OR functionality OR barrier OR Barrierefreiheit OR Nutzbarkeit OR Funktionalität OR Benutzerfreundlichkeit OR Barriere

- Articles in which participants with different disabilities and outcomes were not specifically and separately reported.
- Studies with sighted individuals who were blindfolded, as the focus of the current review was on understanding the first-hand experiences of people with visual impairment and blindness.
- Studies in languages other than English and German.

In the data-charting process, articles were excluded if they did not describe accessibility aspects of digital technologies or if they did not include feedback from the target group on the products. Information described in very general terms was also not documented during data extraction, for example, “software faults were also present” [22], as it was not clear what the faults were. Discussions and authors’ interpretations were not documented, as the review focus was on the reported experiences of visually impaired or blind users.

2.3 Study selection

Titles and abstracts of all articles ($n=683$) were independently reviewed by two reviewers in Rayyan. Regular meetings were held to discuss uncertainties and conflicts, resulting in further refinement of the inclusion and exclusion criteria. In the case of uncertainty or a lack of consensus, the opinion of a third reviewer was sought. In total, 97 articles were selected for a full-text review, of which 39 articles were included in the review. During the full-text review process, any uncertainties were discussed among the authors until an agreement was reached.

2.4 Charting the data

A data extraction table was created in Excel, and the first draft was pilot-tested in six articles by two reviewers independently and further discussed among the authors. An explanation regarding the items was prepared and distributed to all authors [23] to reduce confusion and bias. During the data extraction process, regular meetings were held between the authors to discuss uncertainties, confusion, questions, etc. The key information recorded included the article title, author, year of publication, journal, study location, aim, technology studied, assistive devices used by the participants, participants’ characteristics, tasks, and key results. The table was exported to MAXQDA 2020 (VERBI Software, 2021) for further qualitative analysis.

2.5 Collating, summarizing and reporting results

Several articles followed the WCAG in different ways. Therefore, this information was documented in detail based on POUR elements, as it could provide helpful details for all of the stakeholders in the context of the accessibility of digital technologies for people with visual impairment and blindness. Various checklists were observed and used as references to organize the items; for example, the POUR list was initially prepared using the categorizations of WCAG 2.1 [6] as well as the University of Washington’s accessibility checklist [24]. This information was recorded and, subsequently, further details were added based on the included papers. It is noteworthy that some items, e.g., headings or code validation [24], were not found in any of the selected studies and, therefore, are not listed in the results. The results presented were developed based on the included papers and the reported experiences of the target group in these studies.

Some items are common to several categories; for example, forms (e.g., on websites) need to be designed in an accessible way so that if an error occurs, screen reader users are appropriately informed, so this item can be categorized in the categories “forms” and “input assistance.” However, it was decided to gather all of the errors under “input assistance” to more easily navigate relevant items related to a specific topic. In the example given, the intention was to show how errors should be managed for accessibility and in which areas they need to be considered. Another example is that if a list of links is presented on a website, they have to be accessible via keyboards and shortcuts, so this item could be grouped in both “lists” and “keyboard accessibility” but was grouped only in “lists.” Such categorization is especially helpful in maintaining the focus within a subject (e.g., popups, keyboard), thus helping people without IT expertise to understand and maintain accessibility guidelines.

3 Results

Figure 1 presents the flow of this scoping review. The study characteristics of the included papers are documented in “supplement file 1.” This section explains the POUR items found in the included articles, with more detailed information and examples presented in “supplement file 2.” It should be noted that the results of this scoping review covered some

of the WCAG elements as well as other details that should be considered for the accessibility of digital technologies. User experiences are quoted as examples for clarity, and the article in which the quote was originally used is cited immediately afterward.

3.1 Perceivable

3.1.1 Lists

Different problems with lists were reported by participants in different studies, for example, dealing with a long list to select a date or year [25] or understanding a long list of search results [26], whereas well-placed links [27] and an accessible list of search results [28] were mentioned as facilitators. Therefore, it is recommended that, firstly, either lengthy lists should be avoided or skipping information should be possible for screen reader users to experience easier navigation. Secondly, a group of links should be created as a list with clear labeling, for example, “the menu is a list of links, which were labeled and well described in order to enable the user to navigate through the application without making mistakes” [29]. Thirdly, the list must also be accessible via different ATs, as well as using a keyboard and shortcuts. Shortcut keys were mentioned several times as a means of facilitating the use of long lists for AT users [26, 27, 30]; this is particularly helpful because it allows users to operate without the need for a mouse and scrolling (for more examples regarding lists see [22, 26, 27, 30–32]). Furthermore, a list should be created as a list to allow the screen reader to read it as a list [24], making it easier for the user to understand and navigate.

3.1.2 Layout

Websites and applications usually follow a standard structure so that users know how to work with them or search for the information that they need. When the usual structure (i.e., layout) is not followed, screen reader users could face difficulties in navigating them. Thus, buttons (e.g., menu buttons) should be located consistently and in a size and place that are easily detectable [33, 34]. Important elements should be positioned within reach; for example, if important links are located in the depth of a website, screen reader users will need to listen to all of the content until they achieve their

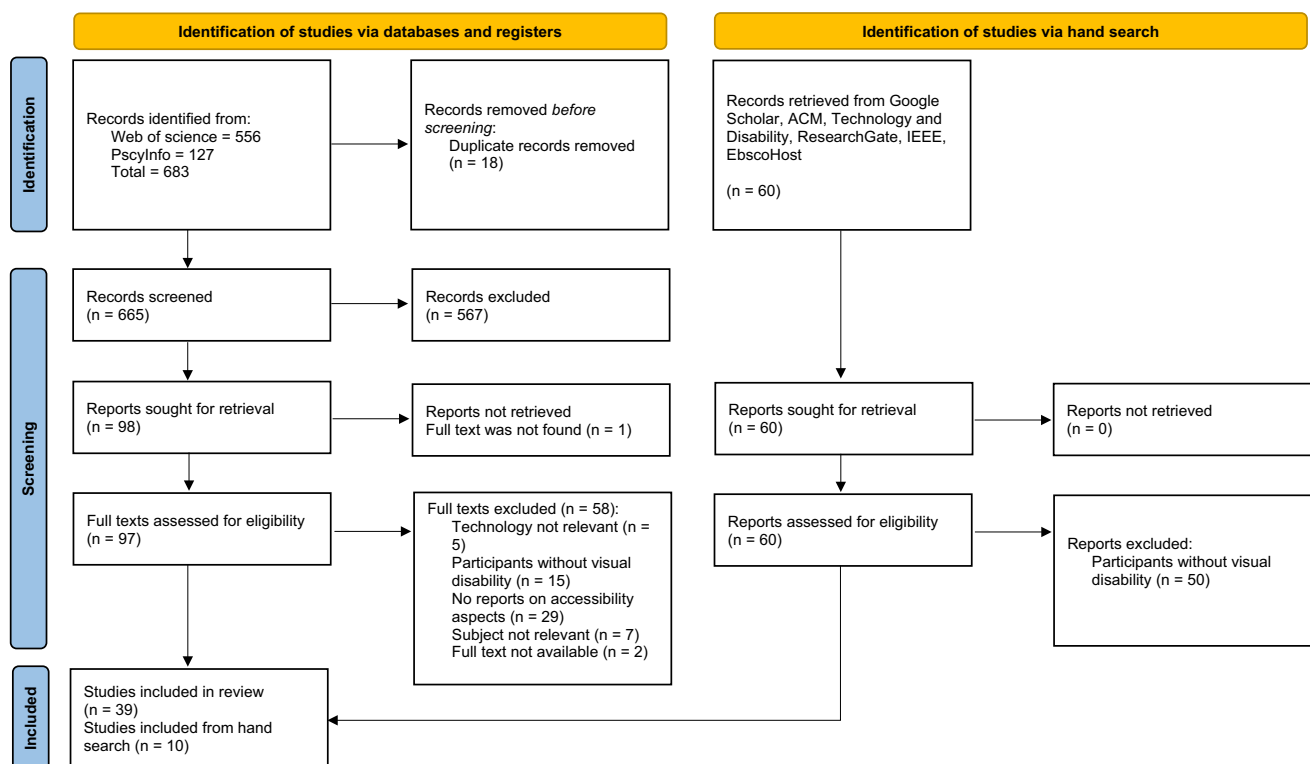


Fig. 1 Scoping review flow

goal [22, 27]. Additionally, participants in Yeong's study found the zigzag text alignment to be inaccessible and preferred a left-aligned format so that information was not lost when using magnification tools. Additionally, they mentioned that they were faced with inaccessibility due to the lack of consistency in text alignment [35]. Furthermore, using a table as the layout should be avoided, as the screen reader will treat it as a table and this could create confusion. Accessible Rich Internet Applications (ARIA) must be used for tables [36] (see Sect. 3.4.3). A simple and less crowded layout is recommended, according to Kamei-Hannan [37]: *"Long reading passages required the test taker to use a scroll bar to view the entire passage. Because of the various fields on the screen, the scroll bar was inaccessible, and a mouse was required to scroll through the text."*

Based on WCAG 1.4.10, content should be presented without a loss of information or functionality, without the need for scrolling in two dimensions (i.e., vertical and horizontal) [25, 38, 39] to allow people with low vision to zoom in so as to enlarge text without losing content. This information was also mentioned in some of the included articles; for example, participants stated that *"zooming/enlarging the page was properly done"* [31] as an indication of accessibility.

3.1.3 Video/audio

Transcripts should be available for videos and audio based on WCAG 1.2.5, but this was mentioned in only one article. In the study by Alajarmeh [25], participants were not able to understand some of the content because the videos did not provide audio descriptions or alt text for the videos.

3.1.4 Forms

A lack of information on the functionality of different elements leads to the exclusion of AT users. For example, Alajarmeh [25] explained that in the case of complex forms or where information is not presented for different functions of a form, participants faced great challenges and experienced inaccessibility. Therefore, forms must be as simple as possible, with clear labels indicating their functionality [40]. Options for easy navigation should be provided, for example, if typing in a field is not possible and the user has to go through a list of options (see lists) [25]. Feedback messages, where the user is informed that their input has (not) been successful, should not disappear in a short time and should be detectable by screen readers [25]. Alajarmeh [25] reported that participants with visual impairment and blindness often face such inaccessibility because the initial coding of programs and applications is not based on the needs of AT users.

3.1.5 Alternative text

Alternative text plays an important role for AT users, particularly those with visual impairment and blindness. Alt text should be provided for graphical and non-textual links [29, 38], images [38, 40], CAPTCHA [41, 42], logos, and non-textual visual content, such as diagrams or charts that present information within an image [36, 37]. For example, in a study of digital tests using ATs, Kamei-Hannan [37] showed that students using ATs could not answer questions when pictures and diagrams did not have alternative text. Singleton and Neuber [36] highlighted that material should be accessible to all types of AT. In their study, although the blind users had no problems regarding the images, as alt text was available, the users who used magnifiers found the PDFs to be inaccessible (as the information was lost when the PDF was resized). In cases of online shopping, Alluqmani [43] reported that a lack of description was challenging; therefore, images of items must offer detailed and clear descriptions. For example, a participant found an item description of *"Take this with you on your summer holiday"* [43] not to be comprehensible, as they could not imagine the meaning of it. In such cases, users have to seek the help of sighted users (for more detailed examples see [43]). Information should not be communicated with visual characteristics, but rather contain alternative explanations that are accessible for AT users [37]. Decorative non-textual content should be marked as decoration so that the screen reader can ignore it (see WCAG 1.1.1).

3.1.6 Tables

Providing facilitators such as shortcuts and accessible navigational links helps AT users to navigate webpages and applications more quickly and more easily. Participants in the study by Singleton and Neuber [36] found it very helpful when PDF or Word files had a linked table of contents. These features help AT users to find content more easily without scrolling or using a mouse [36].

3.1.7 Meaningful sequence

According to WCAG 1.3.2, “when the sequence in which content is presented affects its meaning, a correct reading sequence can be programmatically determined.” Therefore, users need to be informed when a particular sequence needs to be followed. For example, some participants were confused when they had to follow a particular way of choosing dates without receiving any information on it [40]. Sequence disruptions such as popups and advertisements should be avoided, as the user could lose the content on which they were focusing and it could be difficult for AT users to return to it [25]. This shift in focus could be problematic, specifically for screen reader users.

As mentioned previously, any type of accessible navigation link could facilitate the navigation experiences of AT users. In this respect, the use of a heading helps a user to follow the sequence of content, but the content needs to be relevant to its heading level to avoid confusion [44].

3.1.8 Sensory characteristics

According to WCAG 1.3.3, “instructions provided for understanding and operating content do not rely solely on sensory characteristics of components such as shape, color, size, visual location, orientation, or sound.” The reviewed articles emphasized that symbols and special characters must be described in textual format for screen reader users to understand [29, 37].

Additionally, the following elements should be considered: visual clutter on the screen should be avoided [39, 42]; if icons are used, their meanings should be clear to the user [33]; and patterned and transparent backgrounds should be avoided [25, 33]. Visual elements such as color or lines should not be used to communicate information. For example, in the study by Kamei-Hannan [37], students using Braille displays were unable to answer some questions because they relied solely on visual elements (e.g., when answers to questions were based on underlined words that were not recognizable on the Braille display). Moreover, the use of visual elements applies to the use of blank lines (when used for readability purposes), as this causes problems for blind users [22]. For visually impaired users, it is helpful to be able to change the size of text, zoom in and magnify [45] when necessary without losing information (this includes both apps and websites). Fonts and colors used on a website should be accessible and consistent [35]. The color contrast of fonts and backgrounds (neither too low nor too bright) should be considered for people with visual impairment [31]. Alajarmeh [25] reported that the lack of ability to change colors and fonts according to user needs is a severe problem. Thus, it is recommended that, where possible, applications provide the ability to change colors, fonts, etc. [25, 45] so that they can be accessed by different types of AT, for example, on educational platforms [46].

3.1.9 Color contrast

WCAG 1.4.2 details aspects of color contrast and exceptions in detail (see 25, 31, 33, 35, 38 for further information). According to WCAG 1.4.2, the visual presentation of text and images of text should have a contrast ratio of at least 4.5:1. The adjustment of color contrast should be offered so that users can set colors based on their needs. In an example of touchscreens, Huang suggests that such flexibility is specifically helpful for visually impaired users to find information on their touchscreen devices [48]. In a study of an accessible educational platform, Sapp provided users with such adaptability. Such features were positively evaluated by the study’s participants [39], but it should be noted that in Huffman’s study [47], although users were able to adjust the contrast, the highest contrast available was not sufficient for the participants’ needs. This highlights that different needs must be considered in the design and development of technologies.

3.1.10 Text

The font color and its contrast with the background play an important role in the accessibility of digital technologies, and the failure to follow such measures could lead to eye fatigue [33, 39]. A participant in Kim et al.’s study expressed such inaccessibility as follows: “It is difficult to read text because of the unclear color of text. This can easily cause eye fatigue to people with low vision” [33]. Users should be able to adjust the text and font size based on their requirements [25, 32, 39, 45]. Following accessibility measures in detail, such as the text and font size, is very important for an accessible experience for the target group. A participant in a study by Yeong et al. expressed the consistency in text and font size as a positive attribute: “The text all have the same size throughout which is good as I did not have to change my magnification level often” [35] (see WCAG 1.4.4, 1.4.12, and 3.3.1 for further details regarding text accessibility). Such accessibility measures apply to the adaptability of such features based on users’ needs. Users should be able to adjust text sizes without losing

information [31, 39, 48]. For example, a participant in a study by Yeong et al. found such adjustability to be a facilitator: *"I am still able to see the whole page at 200% magnification"* [35].

3.1.11 Popups

As mentioned previously, "screen focus" [41] is important for AT users, as it may not always be easy to return to the original page or content once the focus has been changed. Popup windows change the focus of screen readers, which could potentially cause confusion and disorientation for users [41, 44]. Examples of popups include notifications such as email reminder notifications [41], authentication methods on web applications that cause new screens to pop up [42], or ads [49]. In the case of popups, users should be notified of how to enable (e.g., advertisements), dismiss [24] or handle any form of popup [22, 49].

3.2 Operable

3.2.1 Keyboard accessibility

Content, elements and navigation should be accessible via the keyboard (e.g., tab key and shortcuts) [29, 50, 51] and follow a meaningful sequence. This option allows screen reader users and those using a Braille display to navigate and access websites. Vollenwyder et al. investigated the accessibility of websites in two groups of websites with either low or high compliance with accessibility guidelines, with the low-compliance group highlighting the lack of keyboard accessibility as a problem: *"CTRL + Click worked, but the tab order in their forms is all wrong and the search window is also very keyboard-unfriendly. As expected, pretty bad UX,"* or *"the tab order in the contact form is wrong... when you tab through it jumps from name all the way down and then back up, and the main field can't be tabbed at all"* [31]. In this regard, see also WCAG 2.1 and Sect. 3.1.7. Accessibility through keystrokes and shortcuts provides users with a better experience when using digital technologies, as well as saving time. For example, a participant in a study by Candan et al. [28] found their accessible prototype to be a time saver because it allowed them to perform functions with fewer keystrokes. Another participant also appreciated this specifically for the search results: *"When you clicked on a result, it displayed results in another window so you could just alt-tab to that window, hit alt-/, and you landed right on your target"* [28]. Furthermore, familiar formats should be maintained, as many AT users utilize their memory for navigation (see 3.1.2). This problem was mentioned in a study by Watanabe et al., who explained that the *"first issue occurred when participants were not able to operate the "Select File" push button with the Enter key. The problem was circumvented by pressing the Space key"* [52].

3.2.2 Links and buttons

Links should have sufficient and clear text and buttons clear labels [29, 31, 34, 38, 39] to allow users to understand the purpose of the links and buttons and the destinations to which they will be led, hence facilitating their navigation. For example, blind participants in a study by Carvalho et al. [40] could not complete their tasks due to the *"lack of text for links."* They further explained that *"by clicking of the icons, screen reader read them as "link". Hence, task was not completed as blind users could not find their destination"* [40]. As mentioned previously, visual clutter on a screen should be avoided, especially for visually impaired users who use magnifying ATs. In this regard, the number of links on a page or menu should also be limited [29, 38]. For instance, a participant in Yeong et al.'s study found too many links to be a problem: *"There are too many links to read with a screen reader on Wikipedia"* [35]. Buttons should be easily located [34, 35, 44, 53]; for example, button sizes should not be too small so that visually impaired users can locate them without difficulties [33].

3.2.3 Headings and labels

Headings and their appropriate labels are very important for the accessibility of digital technologies. Navigation using headings with clear labels could save AT users' time and facilitate the use of digital technologies [29, 31, 41]. Headings must be organized in a logical order [35, 53]. For example, a participant in Yeong et al.'s study stated: *"I found it easy to navigate through the different headings for each page using the 'Insert-F6' function on JAWS which lists all the headings"* [35]. In addition, headings and page titles should represent the content intended. For example, *"in many cases, the content appeared to be visually divided into heading levels; however, the underlying representation of the content was not implemented"*

using structural heading levels. This caused problems in navigating over the content, especially on mobile Apps” [34]. For more information see WCAG 2.4.6.

3.2.4 Navigation

Navigation was one of the most frequent items mentioned and in included articles. Here some of important factors will be reported. Further detailed information is presented in “supplementary file 2.”

Each page should clearly indicate its content for screen reader users to know the purpose of a page and navigate it easily; for example, having clear page titles could facilitate navigation using ATs [34, 38, 40, 54]. Options to facilitate navigation should be offered, for example, “jump to top buttons” at the end of long webpages [35] or the correct sequences of actions. A blind participant in Zeinullin and Hersh’s study [55] found the lack of clarity in navigation to be a problem: *“I wish the app guided me while I was exploring images. Like, in which direction I should move in order to get to the object”* [55]. Icons, links and items should be located in places that AT users can find easily [25, 26, 28, 31, 33, 34, 39, 42, 44, 46, 53, 56], for example, finding products easily on online shopping websites [31] or attachment buttons on an educational platform [46].

One of the frustrating problems that users faced was being stuck in a loop, as a clear exit from a webpage was not detectable [38, 53]. This also applied to cases in which by clicking on an unwanted link or button, they could not navigate back to where they were; for example, popups could confuse screen reader users’ navigation [42]. For further examples see [25, 26, 49]. It is helpful when important navigational links are easily detectable to all users [25, 27, 35]. For example, when they are located in menus, less scrolling should be offered (drop-down menu [35]) or positioned strategically in the layout, where it is easily detectable for AT users [39]. Keyboard accessibility for navigation is an important factor [50], where the consistency of keystrokes should be followed; for example, a participant in a study by Vollenwyder et al. stated: *“In all input fields you could navigate with the tab key (first name, last name, street etc.), but in the last field Your message you could not get to it with the tab key you have to search for the field explicitly”* [31].

The studies also frequently mentioned the search and results function. Search results should have clear and sufficient labels and headings so that screen reader users can understand and navigate them [29]. Participants in a study by Vigo and Harper [38] were confused because their search did not have any results. In such cases, users should be clearly informed when no search results are found [38, 54]. In addition, search results should be structured clearly so that screen reader users can understand and navigate them [22, 29, 31, 38]. Organizing long search results into clusters was found to be helpful and time-efficient by participants in the study by Aqle et al. [57], who tested an accessible search engine prototype. Therefore, it is recommended that when too much information or too many links are found, a cluster of search results should be offered to help screen reader users. In this example, the screen reader user can skip the unwanted clusters, rather than read out each result aloud, which could cause confusion or be time-consuming: *“This is very helpful to categorize search results”* [57]. Mobile apps with a large amount of content should offer a search function so that the user does not have to scan the content completely [25]. See WCAG 2.4 and “supplement file 2” for further information and examples regarding navigation.

3.2.5 Auto-updating content

Carousals create difficulties for screen reader users because the information that they provide is automatically updated and the user has no control over their speed of occurrence. This issue is specifically challenging for screen reader users, as this fast change of focus could lead to a loss of information as well as confusion [25]. Both blind and visually impaired participants in Alajarmeh’s study [25] found auto-updating content to be a serious problem. In this regard, Yeong et al. concluded that time-sensitive carousel content could lead to user confusion and frustration, particularly for those who use ATs [35]. For further examples see [35, 38, 48] and WCAG 2.2.2 and 2.2.4.

3.2.6 Time limit

If a form, application, learning platform, etc. has a time limit, the user should be made aware of this and allowed to postpone or extend the time limit [24]. In cases of a malfunction of a timed task or process, the user should be able to find their way back as easily as sighted users. This issue occurred in the study by Muwanguzi and Lin [58], where a participant needed two hours to retake a timed test, whereas a sighted user required only 15 min. Errors in timed tasks were also

mentioned in Alajarmeh's study [25], where participants had to repeat the process. Such errors cause AT users to invest more time in completing their tasks. See WCAG 2.2.1, 2.2.5 and 2.2.6 for more details.

3.2.7 Flashing and flickering content

In a study by Vollenwyder et al. [31], users found a lack of flashing content to be a positive attribute, but no other study included in this review has mentioned this item.

3.3 Understandable

3.3.1 Language

In several studies, complicated and technical information was a challenge for the target group. For example, Nimmolrat et al. [59] reported that visually impaired users avoided some applications because the information provided was difficult to understand. In a usability evaluation study of web-based tourist information software, Stary and Totter [50] reported that the target group considered well-explained information to be an important factor and concluded that content must be developed based on the needs of the target group. Therefore, clear, readable and understandable language for the target group should be used [31, 33, 35, 43, 50, 59, 60]. Unexpected and unfamiliar characters should be avoided [38]. Acronyms should be explained in the context. A lack of such information could specifically cause confusion and difficulties for screen reader users following the content, as screen readers treat acronyms as a word: *"For example, the acronym ACM is incomprehensibly read as one word that confuses the user"* [25].

3.3.2 Predictability

WCAG 3.2 states that webpages should appear and operate in predictable ways. For example, unexpected changes or functions like banners [38] could cause difficulties for users.

Usually, specific links, icons or features are positioned as a standard in certain places on websites or apps, with users eventually learning these to develop a mental model and, consequently, looking for them in their designated places. For example, a participant in a study by Vollenwyder et al. criticized the fact that the contact details were not placed where they usually are, either in the footer or in the menu [31]. In this regard, an unfamiliar layout, icons, buttons, links and features should be avoided [33, 38, 40]. When these standards are not followed, users are challenged to navigate or accomplish their goals. In contrast, Craven et al. [27] reported that visually impaired users outperformed sighted users in some cases when navigation links were positioned appropriately.

Users expect a particular answer by performing specific actions. Yeong et al. [35] noted that predictable navigation is an essential part of accessibility. In their case, participants were confused when using their prototype because the search and results function was not similar to the common search engines to which they have used. When the search function of a website or app is used, users expect to obtain a list of results relevant to their search. If the focus thereafter is not on the search results, the screen reader user could face difficulties in navigating (see [25, 31, 35, 38] for examples).

3.3.3 Input assistance and errors

WCAG 3.3 advises that users should receive support to avoid and correct mistakes. Sighted users usually find it easier to rectify errors than do blind users, according to Shimomura et al. [22]. Alnfai and Sampalli [42] confirmed this in their study in which participants' mistakes, here entering information in the wrong places and failing to log in, led to confusion and time spent on correcting these errors (for more examples see [22, 42, 50]). In this regard, when an action is performed, users should receive accessible feedback regarding their action, allowing them to understand the result of their action. For example, for online shopping experiences, a participant in a study by Vollenwyder et al. stated that: *"Although I could select all products, only in the shopping cart I have not brought them. Each time came: Your shopping cart is empty. There was no feedback that a product was assigned to the cart, nor a button, add to cart or anything else"* [31]. In a study of a prototype smart home system, de Oliveira et al. [34] explained that accessible feedback plays an important role for visually impaired users. Feedback allows users to know whether their actions have produced the expected response, for example, whether or not the lights have been turned off. In this regard, WCAG 3.3.1 indicates that users should be aware that an error has

occurred and can determine what went wrong. The error message should be as specific as possible and errors should be consequently described in a textual form for screen reader users [25, 44].

3.4 Robust

3.4.1 Compatible

WCAG 4.1 suggests enabling compatibility with ATs. Moreover, several studies have mentioned the importance of websites and applications being compatible with different types of AT [40, 51, 61]. On the positive side, where such compatibility provides user satisfaction and the user's ability to use digital technologies, participants in a Yeong et al. study stated: *"The website works really well and is really fluid when using Voiceover on my iPhone,"* or *"I found the website easy to understand and JAWS friendly"* [35]. However, when compatibility with different ATs is not offered, AT users are challenged and sometimes avoid using digital technologies. For example, a participant in a study by Alnfai et al. expressed: *"I do not open the bank account from the phone because I feel it is not safe and the web application is not accessible because it does not support the VoiceOver service"* [42]. Such difficulties were also observed on tablets and smartphones in a study by Leporini and Buzzi [32], which explored the use of eBooks by the target group. A participant stated: *"Many editing functions are not available or are impracticable via screen reader and gestures on the touch-screen"* [32]. Mobile apps, including browser apps, should be accessible for ATs as well, for example, for visually impaired users who need to magnify the screen [45]. Another example is CAPTCHA or images that should be accessible to the screen reader, and content should be accessible via different alternatives [49]. In cases of the inaccessibility of CAPTCHA, e.g., an unclear voice [42] or inaccessibility to screen readers [49], users have no choice but to seek sighted help [25].

In cases of educational platforms or online tests, compatibility plays an important role in the inclusion of AT users. For an example of challenges caused by underlined words and the lack of compatibility with Braille, see Sect. 3.1.8 and [37].

3.4.2 Dynamic content

There was an overlap of meaning in the categories of "auto-updating content" and "dynamic content." Several articles considered dynamic content to be content that plays automatically at short intervals and over which the user has less control. However, dynamic content could also mean content that changes based on the user's data and preferences. Examples include advertisements on websites or the way in which websites such as YouTube work. Most comments used the definition of the former. Therefore, we included all such comments in "auto-updating content," even though users utilized the words "dynamic content."

Regarding dynamic content, AT users rely usually on their memory and build mental models of how a website or app functions. Dynamic content (such as ads) that causes changes based on users' behavior could create difficulties for users who rely on their mental models [56].

3.4.3 Accessible Rich Internet Applications (ARIA)

ARIA can be added to HTML so that screen readers and other types of AT can better understand the elements on a webpage, and can be used to access dynamic content or page regions [24]. Buzzi and Leporini [29] mentioned that the use of ARIA could have increased the accessibility of Wikipedia for screen reader users, improving navigation and comprehension (see Sect. 3.1.2).

4 Discussion

This scoping review focused on the accessibility of digital technologies for people with visual impairment and blindness, identifying items facilitating their use based on first-hand experiences of the target group and the existing accessibility guidelines. The studies included in this review covered a variety of contexts, including work, education, and daily activities such as online shopping.

It was noted that users would like to be able to make changes based on their needs, for example, in terms of color contrast, text font, text color, and text size [39]. This flexibility is particularly helpful for people with visual impairment, for example, when they use magnifier applications to access online platforms and websites. Additionally, font shape, the

space between letters and lines, and stroke width were identified as factors that influence the readability of text, and that consequently might contribute to eye fatigue: *"It is difficult to read text because of the unclear color of text. This can easily cause eye fatigue to people with low vision"* [33]. This highlights the importance of being able to customize the details of apps and websites to suit one's needs and preferences. It is therefore advisable to offer such flexibility, especially in mobile applications, so that customization is possible.

In this review, no limitations were considered for the year of publication to detect any patterns. It was observed that some accessibility issues remain consistent. For example, keyboard accessibility on websites seems to have been a problem for screen reader users from as early as 2004 [27] to 2022 [49]. Following a universal design perspective when developing new technologies could prevent such limitations that all people with disabilities face while using technologies.

It was repeatedly mentioned that communication using visual elements must be avoided on websites, apps, social media, and so on. Using images as icons or a way of communication without alternative text makes such tools inaccessible for screen reader users. The role of each element, e.g., UI elements, must be clear [25]. The fact that many websites and applications are designed for sighted users was also reported in many studies; for example, users with blindness could not use an online banking website, as the information on the homepage was highlighted visually, so while sighted users could find their way on this website, users with blindness were excluded [26].

Compatibility with different ATs must be taken into account. Singleton et al. [36] showed that even though an accessible PDF file (e.g., containing alt text for images) for screen reader users was easy to navigate, visually impaired users who use screen magnifiers were challenged because by zooming in, the text, images and columns would collide and the users would lose some information, affecting their educational performance. In this case, measurements could be taken to avoid such limitations (text size). Additionally, CAPTCHA should be accessible for users of ATs; for example, the audio version of text must be clear for users [42]. In such cases of inaccessibility, users have to either abandon using the technologies or seek assistance from those without visual impairment [25].

In online shopping when sighted users look at images to understand the details and make decisions, this is challenging for blind users because the descriptions provided are usually short and do not provide enough information regarding how they look. For example, a participant in a study by [43] stated: *"Quite often, it will just say blue and white, but it wouldn't tell you [if] it's blue and white stripes or blue with white flowers or, you know, blue and white spots... I don't know which one is most... which one has the most color."* Furthermore, some users stated that they ignore images on websites altogether because, based on their experience, they never are accessible: *"I ignore images on a website as they've never had useful descriptions"* and *"images do not affect shopping experience as alt-text is normally not good"* [60]. Users claimed that, instead of hearing the alt text of images, they usually read the review comments from users who have already purchased these items because they provide detailed information such as the color, if they are formal, etc. In this regard, Tigwell et al. [62] explored the use of emojis from the perspectives of people with visual impairment and blindness. While sighted users might benefit from using these features easily, screen reader users could be challenged. Firstly, some emojis do not have appropriate alt text, confusing screen reader users. For example, *"Email subjects have emoji now; Ebay put a [truck emoji] to show your order has been sent. For a long time, I [was] puzzled as to why they'd write the word 'truck' there."* Secondly, searching the long list of emojis and finding the one that they need is laborious: *"...finding the right one to send. I either don't know whether it exists or what it is, or where to find it. Sighted people just glance at a screen and can find them pretty quickly, while we have to go through all of them."* Lastly, for some blind users, comprehending the meaning of such emojis is not easy, as without eyesight or assistance their use is not possible: *"Some emoji [are] useless or just have a bad design (I was told the 'pray' emoji ... is actually a 'high five'; or 'Emoji is something fun for sighted texters... but for me it's just an extra string of words. ...like the grinning face emoji; it looks fun and cute when you look at it, but Voiceover describes it as 'grinning face with clenched teeth emoji' which sounds more like a grimace than a big smile."*

Individual differences in the topics of disability are widely observed and the accessibility of digital technologies is also to some extent influenced by individual differences. For example, color contrast preferences vary among individuals: *"most participants preferred a black background with white text, but one participant preferred black background with yellow text"* [63]. Another example of individual differences was detected in the preference for using menus. While the participants in the study by Yeong et al. [35] found drop-down menus to be helpful (as less scrolling is required), participants in the study by Sapp et al. [39] were happy to have alternative possibilities to drop-down menus.

Regarding accessibility, even though guidelines offer constructive and helpful frameworks, they do not suffice for complete accessibility of digital technologies for people with disabilities [25, 31, 54] and there remains room for improvement. This scoping review highlights details of first-hand experiences of the target group, and such information is a helpful source for improving accessibility. In addition, prototypes that are focused on improving digital accessibility provide constructive details that could be transferred to the development and design of accessible digital technologies. However,

this type of information is usually lost in the pool of available information. Therefore, this review contributes to bringing together all of these attempts so that all stakeholders may benefit from existing research. The examples provided in this review, as further explanations of WCAG elements, can also help all stakeholders, even those with no or less IT knowledge, to better understand accessibility and try to implement it in their products. Despite all of these benefits, this review has some limitations. Firstly, the review only focused on one type of disability, so further research is needed that focuses on different types of disabilities. In addition, other databases could be considered to include other publications.

This review's findings are consistent with those of Khan & Khusro's literature review [12] on smartphones, as they highlight influential factors such as the importance of meaningful sequences, accessible navigation, compatibility with other ATs, consistency, and flexibility. However, where this review provides a more detailed description of items based on the target group's experiences, Khan and Khusro's review [12] offers a summary of the importance of each factor. Furthermore, this review is in line with other studies in that accessibility guidelines need to be improved. For example, on the topic of the accessibility of mobile applications, Di Gregorio et al. [3] argued that, although there has been increased interest in accessibility and the development of ATs, there is room for improvement in accessibility guidelines for mobile applications. Secondly, in agreement with Ashraf et al. [64], all stakeholders, including technology developers, need to be informed about these guidelines and asked to implement them in earlier stages of design and development. It is noteworthy that previous studies and reviews have predominantly concentrated on a specific technology or certain features of technologies; for instance, Shera et al. [65] developed a checklist for the user interface of mobile applications based on the accessibility issues of the target group. This review, however, encompassed a broader range of technologies, focusing on accessibility and identifying both challenges and facilitators. In conclusion, accessibility starts from the moment at which technologies are designed and developed to the moment at which they are implemented. Consequently, it is important not only to educate product and application developers about accessibility but also to train AT users to use these technologies competently. A checklist will be developed based on the review findings, further analysis of the results, and in consultation with the target group. This checklist is intended to be a tool for stakeholders in accessible development and use of digital technologies.

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Declarations

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